

Cell Biology Notes

Photosynthesis occurs in the chloroplasts of plant cells. Chlorophyll absorbs light energy, which is used to convert carbon dioxide and water into glucose and oxygen.

Mitochondria are the organelles that produce most of the cell's ATP through cellular respiration. They have a double membrane and their own circular DNA.

The cell membrane is a phospholipid bilayer. It is selectively permeable and controls which substances enter and leave the cell. Osmosis is the diffusion of water across this membrane.

Genetics

DNA replication is semi-conservative: each new double helix contains one original strand and one newly synthesised strand. The enzyme DNA polymerase adds nucleotides in the 5' to 3' direction.

Gregor Mendel studied pea plants and formulated the law of segregation and the law of independent assortment. A dominant allele masks the effect of a recessive allele.